Simulating Mechanical Transfer in Rowing

From Thrust–Drag Energy Balance to Disturbance–Efficiency Coupling via Multibody Dynamics and IMU-Informed System Identification

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Abstract

Rowing measurement culture is exceptionally good at quantifying *power*. Yet elite performance depends at least as strongly on the *cost of power*: losses induced by velocity fluctuations, hull attitude dynamics (pitch/trim), and recovery-phase mass motion that increase effective drag and reduce how much propulsive work becomes uninterrupted run. This paper unifies two studies into a single, publication-oriented modeling framework.

First, we reproduce a baseline thrust–drag model of an eight (8+) using a periodic drive/recovery forcing and quadratic drag. The baseline provides an analytically transparent energy accounting in which the large majority of rowers' mechanical work is dissipated by drag, and it reproduces characteristic within-stroke velocity oscillations and their sensitivity to drag, mass, and stroke rate. Second, we extend the baseline in two directions that specifically expose technique-dependent loss channels: (i) a reduced-order multibody model that includes crew center-of-mass motion on the slide and a pitch surrogate coupled to drag, and (ii) an IMU-informed analysis pipeline (orientation estimation, gravity removal, stroke segmentation, time–frequency features) plus a grey-box system identification formulation to estimate disturbance-sensitive parameters from accelerometer/gyro signals.

The combined study reframes performance optimization as a loss-minimization problem: reducing disturbance-induced drag augmentation and velocity-check penalties can deliver meaningful speed gains without increasing peak power. We conclude with forward-looking methodological pathways—validated IMU datasets, coupled CFD/ROM drag models, and program-scale system identification—toward measuring and minimizing technique-driven losses before on-water seat trials.

Keywords: rowing; hydrodynamic drag; velocity fluctuations; multibody dynamics; inertial measurement units; system identification; efficiency; stroke segmentation.